



ResearchLens Analysis Results - Complete Guide



What is ResearchLens?

ResearchLens is an intelligent research analysis system that automatically reads, understands, and analyzes scientific papers. Think of it like having a super-smart research assistant who can:

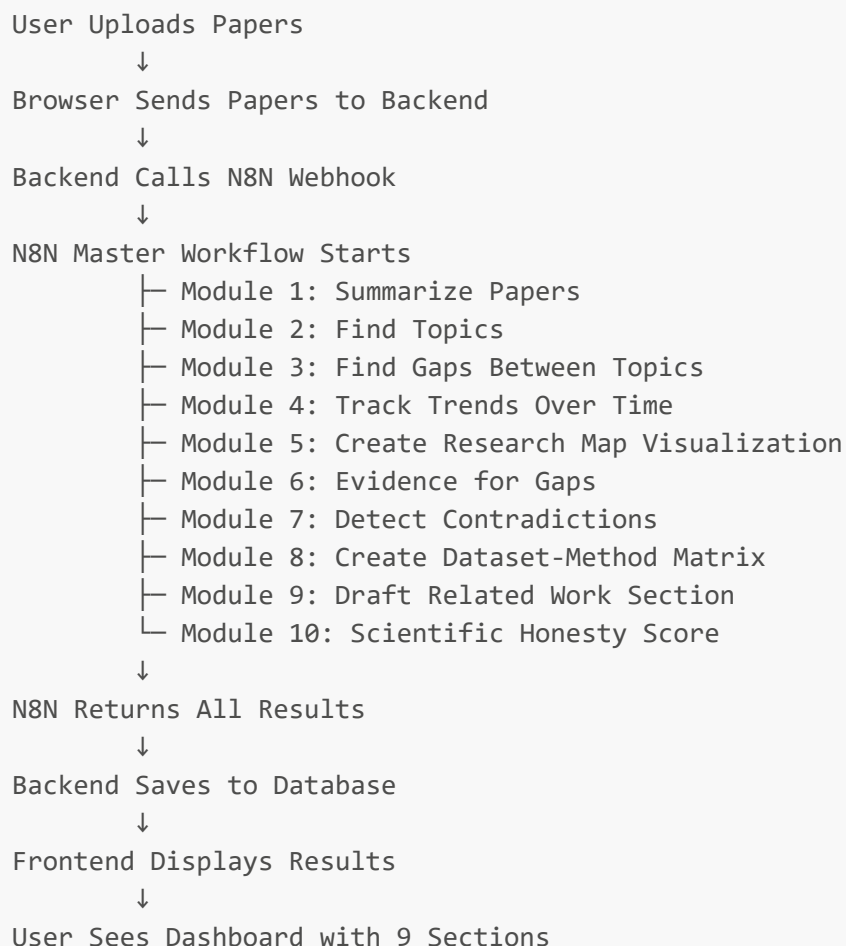
1. **Read** dozens of papers and summarize each one
2. **Find patterns** across all papers (topics, themes, methodologies)
3. **Spot gaps** where research hasn't connected different topics yet
4. **Track trends** to see which topics are growing or declining
5. **Create visualizations** to show how research areas relate to each other
6. **Write reports** with AI reasoning about what the gaps mean



How N8N Orchestrates Everything

N8N is an automation platform that coordinates all the analysis steps. Instead of running modules one-by-one on your computer, N8N runs them in parallel across a dedicated workflow:

The N8N Workflow Flow



Why use N8N instead of running modules locally?

-  Faster (modules can run in parallel)
 -  More reliable (N8N handles errors and retries)
 -  Scalable (can handle bigger datasets)
 -  Separated concerns (frontend doesn't do heavy computation)
-

The 10 Analysis Modules Explained

Each module is like a specialist researcher focusing on one specific task:

Module 1: Summarization

What it does: Reads each paper and extracts the most important sentences

Simple analogy: Like reading a book and writing its summary on a post-it note

How it works:

1. Splits each paper into sentences
2. Extracts keywords that appear most frequently
3. Scores each sentence based on how many keywords it contains
4. Keeps the top 3 sentences = summary for that paper

Results stored:

```
{
  paperId: "paper_123",
  title: "Deep Learning in Computer Vision",
  summary: "Three most important sentences from the paper",
  keywords: ["deep learning", "neural networks", "image classification", ...]
}
```

Simple metrics:

- **Count:** How many papers were summarized (usually = number of input papers)
-

Module 2: Topic Modeling

What it does: Groups papers by their main topics/themes

Simple analogy: Like sorting books in a library into different sections (Science, History, Fiction, etc.)

How it works:

1. Takes all paper summaries and keywords
2. Uses machine learning to find groups of similar papers
3. Assigns a semantic label to each group (the topic)
4. Uses Ollama AI to generate human-readable names

Results stored:

```
{
  id: "topic_1",
  name: "Machine Learning and Neural Networks",
  keywords: ["deep learning", "neural networks", "training"],
  coherence: 0.85, // How well papers in this topic "fit together"
  paperIds: ["paper_1", "paper_5", "paper_12"], // Which papers belong here
  llm_label: "AI/ML Foundations", // AI-generated label
  llm_domain_summary: "Core algorithms for neural computation",
  llm_confidence: 0.92
}
```

Metrics you'll see:

- **Coherence** (0-1): How similar the papers in a topic are to each other. Higher = better.
- **Paper Coverage**: How many papers belong to each topic

Module 3: Gap Detection 🔍

What it does: Finds missing connections between different topics

Simple analogy: Like noticing "Wait, why do Topic A (Computer Vision) and Topic B (Natural Language Processing) never talk to each other? They could learn from each other!"

How it works:

1. Takes all topics from Module 2
2. Calculates similarity between each pair of topics
3. Counts how many papers mention both topics together (co-occurrence)
4. Topics that are similar BUT don't co-occur together = a gap
5. Uses Ollama AI to explain what the gap means

Results stored:

```
{
  gapId: "gap_1",
  topicA: "topic_1",
  topicB: "topic_2",
  topicAName: "Machine Learning",
  topicBName: "Knowledge Graphs",
  similarityScore: 0.72, // How related these topics are
  coOccurrence: 0, // How often they appear together
  gapScore: 0.85, // How significant this gap is (0-1)
  explanation: "ML models and knowledge graphs rarely discussed together, despite both being AI paradigms",
  llm_gap_significance: "high",
}
```

```
llm_integration_opportunity: "Semantic reasoning in neural models"
}
```

Metrics you'll see:

- **Similarity Score** (0-1): How conceptually related two topics are
- **Co-occurrence**: Number of papers mentioning both topics. 0 = never mentioned together!
- **Gap Score** (0-1): Importance of this gap. Higher = bigger opportunity for research

Module 4: Trend Detection

What it does: Tracks which topics are growing, stable, or declining over time

Simple analogy: Like checking stock market graphs - which research areas are hot right now?

How it works:

1. Groups papers by year
2. Counts how many papers about each topic in each year
3. Detects patterns: Rising (more papers each year), Stable (same), Declining (fewer papers)
4. Uses Ollama AI to explain what the trend means

Results stored:

```
{
  topicId: "topic_1",
  topicName: "Deep Learning",
  trend: "rising", // or "stable" or "declining"
  growthRate: 0.45, // How much is it growing
  peakYear: 2024, // Best year for this topic
  yearlyCounts: [
    { year: 2020, count: 15 },
    { year: 2021, count: 28 },
    { year: 2022, count: 42 },
    { year: 2023, count: 68 }
  ],
  llm_trend_summary: "Deep learning research is accelerating",
  llm_paradigm_shifts: ["From supervised to self-supervised learning"],
  temporalConfidence: 0.88
}
```

Metrics you'll see:

- **Trend**: Rising, Stable, Declining, or Insufficient Data
- **Growth Rate**: How much the topic is growing
- **Peak Year**: When this topic had the most papers

Module 5: Visualization (Research Map)

What it does: Creates a visual map showing where papers are positioned and how topics relate

Simple analogy: Like a map of a city where each point is a paper, papers in the same topic cluster together, and lines show connections between topics

How it works:

1. Reduces paper data to 2D coordinates using UMAP (a dimensionality reduction algorithm)
2. Plots each paper as a point on the research map
3. Colors points by topic (Machine Learning = blue, NLP = green, etc.)
4. Draws curved lines connecting related topics

Results stored:

```
{
  map: {
    points: [
      {
        paperId: "paper_1",
        title: "Deep Learning Fundamentals",
        x: 150, // Position on canvas
        y: 200,
        topicId: "topic_1",
        topicName: "Deep Learning",
        year: 2023
      },
      // ... more points
    ],
    links: [
      {
        topicAId: "topic_1",
        topicBId: "topic_2",
        strength: 0.65 // How related they are
      },
      // ... more links
    ]
  }
}
```

What you see in the dashboard:

- **Points:** Each is a research paper. Hover to see paper details
- **Colors:** Different colors = different topics
- **Clusters:** Papers bunched together = same topic
- **Curved lines:** Connections between topics (stronger = thicker lines)

Module 6: Gap Evidence

What it does: Finds specific papers that could bridge the gaps

Simple analogy: "We found a gap between Topics A and B. Here are some papers that mention both topics - they could be bridges!"

How it works:

1. Takes each gap from Module 3
 2. Searches for papers mentioning both topics
 3. Uses Ollama AI to verify they're actually bridging papers
 4. Explains how each paper connects the gap
-

Module 7: Contradiction Detection ⚠️

What it does: Finds papers with conflicting conclusions

Simple analogy: "Paper A says method X is best, but Paper B says method Y is better"

How it works:

1. Looks for papers in same topic with opposite claims
2. Uses Ollama AI to identify contradictions
3. Explains what the conflict is about

Results stored:

```
{
  contradiction: {
    paperId1: "paper_5",
    paperId2: "paper_12",
    claimA: "Model A achieves 95% accuracy",
    claimB: "Model A only achieves 87% accuracy",
    explanation: "Different datasets or evaluation metrics used"
  }
}
```

Module 8: Dataset-Method Matrix 📋

What it does: Creates a table showing which datasets are used with which methods

Simple analogy: Like a spreadsheet showing "Machine Learning Method X has been tested on Datasets A, B, and C"

Results stored:

```
{
  matrix: {
    datasets: ["ImageNet", "COCO", "CIFAR-10"],
    methods: ["CNN", "ResNet", "Vision Transformer"],
    usage: {
```

```
"ImageNet": ["CNN", "ResNet"],
"COCO": ["CNN", "Vision Transformer"],
// ...
}
}
}
```

Module 9: Related Work Draft 📝

What it does: Automatically generates a "Related Work" section for a research paper

Simple analogy: Like writing the background section of a paper that discusses all the papers you studied

How it works:

1. Groups papers by topic
2. Orders them chronologically
3. Uses Ollama AI to generate narrative paragraphs connecting them
4. Creates a polished "Related Work" section

Results stored:

```
{
  relatedWork: {
    markdown: "# Related Work\n\n## Deep Learning Foundations\n...",
    sections: [
      { topic: "Deep Learning", papers: [...], narrative: "..." },
      // ...
    ]
  }
}
```

Module 10: Scientific Honesty Score ✅

What it does: Analyzes if papers properly cite related work and avoid plagiarism

Simple analogy: Like a teacher checking if a student properly cited all sources in their essay

How it works:

1. Checks citation patterns
2. Uses Ollama AI to verify claims are properly supported
3. Calculates a score (0-1) indicating research integrity

Results stored:

```
{
  honestyScore: 0.87, // 0-1 scale
  citationCoverage: 0.92, // How many related works are cited
  claimSupport: 0.85, // How well claims are backed by evidence
  analysis: "Generally good research practices with minor citation gaps"
}
```

The 9 Dashboard Sections (What You See)

After all modules run, the frontend displays results in 9 sections:

Section 1: Dataset Summary

Location: First section in results

What it shows:

- Total number of papers analyzed
- Number of topics found
- Number of gaps identified
- Year range of the papers
- Quality metrics calculated from topic coherence, coverage, and gap novelty

Metrics explained:

Total Papers	Topics	Gaps	Years
145	12	34	20-2024

Quality Metrics (0-1 scale):

- Topic Coherence: How well papers in each topic fit together
- Topic Coverage: What percentage of papers are assigned to topics
- Gap Novelty: How significant are the gaps found
- Model Quality: Overall quality score (combination of above)

Why these matter:

- More papers = deeper analysis
- More topics = more diverse research
- More gaps = more opportunities for new research
- High coherence = reliable topic detection

Section 2: Analysis Summary

Location: Second section after dataset summary

What it shows:

- Executive summary of findings (auto-generated by Ollama AI)
- Key findings from the analysis
- Main insights and takeaways

Example summary:

This analysis of 145 papers in AI/ML research (2020-2024) identified 12 major research topics with 34 significant gaps. Key findings include:

1. Deep Learning is rising (68% more papers in 2024 vs 2020)
2. Knowledge Graphs and NLP rarely collaborate (major gap)
3. Contradiction detected: Different accuracy claims for ResNet

Section 3: Gap Detection Results

Location: "Main Output" section showing research opportunities

What it shows:

- Ranked list of all gaps found (most important first)
- For each gap:
 - Name: "Topic A × Topic B"
 - Similarity Score: How related the topics are
 - Co-occurrence: How often they appear together
 - Gap Score: How important this gap is

How to read a gap card:

RANK #1: Machine Learning × Knowledge Graphs

Similarity: 0.72 (Very similar topics)

Co-occur: 0 (Never together!)

Gap Score: 0.85 (Major opportunity)

Explanation:
ML models and knowledge graphs are both AI paradigms but rarely discussed together despite potential synergies. This represents a significant research opportunity in semantic reasoning.

Click "View" to see papers that could bridge this gap

Interactive features:

- Click any gap to see details
- View papers that mention both topics

- See AI reasoning about why the gap exists

Section 4: Trend Analysis Results

Location: Shows how topics are growing/declining

What it shows:

- Summary: How many topics are rising, stable, declining
- Charts showing papers per topic over time
- Trend type for each topic

Reading the trends:

TOPIC: Deep Learning



Papers mentioning "deep learning":

2020: 15 papers
2021: 28 papers
2022: 42 papers
2023: 68 papers

What trends mean:

- **Rising** (↗): This area is hot! More researchers working on it
- **Stable** (→): Established field, consistent interest
- **Declining** (↘): Less popular now, possibly mature or replaced by newer topics

Section 5: Research Map

Location: Visual representation of all papers and topics

How to read the map:

Each DOT = one research paper
Dots GROUPED by COLOR = same topic

Machine Learning ●●●●●
(Blue dots close together)

NLP ●●●●

(Red dots close together)

← Papers mentioning both topics are positioned between the groups
Curved LINES between colors = topics are related

Interactive features:

- **Hover over a dot:** See paper title and topic
- **Click a dot:** See full paper details (title, year, main findings)
- **Hover over a line:** See which topics are connected
- **Color legend:** Shows all topics and their colors

What the positions mean:

- **Close dots** = papers very similar to each other
- **Spaced out dots** = papers somewhat different
- **Gaps in the map** = research areas not well covered
- **Lines connecting colors** = topics that reference each other

Section 6: Chatbot

Location: Q&A interface about the analysis

What it does:

- Ask questions about the analysis results
- Examples:
 - "What are the top 3 gaps in this research?"
 - "Which topics are rising the fastest?"
 - "Are there contradictions in the data?"
 - "What papers bridge gap #5?"

How it works:

- Uses Ollama AI (local LLM)
- Answers based on analysis results
- Cites specific papers and gaps

Section 7: Scientific Honesty

Location: Shows research integrity metrics

What it shows:

- Honesty score (0-1)
- Citation coverage: How many related works are cited
- Claim support: How well are claims backed by evidence
- Potential plagiarism detection
- Missing citations warnings

Example:

SCIENTIFIC HONESTY SCORE: 0.87/1.0 ✓

Citation Coverage: 92%

└ How many related works are cited

Claim Support: 85%

└ How well are major claims backed by evidence

Issues Found: 2

- Paper #5 claims not well cited
- Paper #12 possible citation gap in literature review

Section 8: Dataset-Method Matrix

Location: Table showing method-dataset combinations

How to read it:

	CNN	ResNet	Vision-T	Transformer
ImageNet	✓	✓	✓	
COCO	✓		✓	✓
CIFAR-10	✓	✓		
Pascal	✓			✓

What this means:

- ✓ = This method tested on this dataset
- Empty = Never tested together (potential gap)
- X marks show research coverage

Why it matters:

- Shows which combinations are well-researched
- Highlights untested combinations (research opportunities)

Section 9: Related Work Draft

Location: Auto-generated related work section

What it is:

- Ready-to-use content for your research paper
- Organized by topic
- Chronologically ordered papers
- AI-generated narrative connecting them

Example structure:

```
# Related Work

## Deep Learning Foundations (2020-2022)
[3-4 papers grouped, narrative connecting them]

Deep learning has emerged as a fundamental paradigm...
[AI-generated text citing and explaining papers]

## Knowledge Graphs and Semantic Understanding (2021-2024)
[Papers organized chronologically]
...

## Emerging Trends: Multimodal AI (2023-2024)
...
```

How to use it:

- 1. Copy the entire section into your paper
- 2. Review for accuracy (always verify AI-generated text!)
- 3. Add your own insights
- 4. Cite all papers properly

 Key Metrics Explained

When you see numbers, here's what they mean:

Metric	Range	What it means
Coherence	0-1	How well papers in a topic fit together. 0.8+ is good
Topic Coverage	0-1	What % of papers are grouped into topics. Higher is better
Gap Score	0-1	How important is this gap. 0.7+ is significant
Similarity Score	0-1	How related are two topics. 0.7+ = very related
Co-occurrence	0+	How many papers mention both topics. 0 = never together
Growth Rate	%	How much a topic is growing per year
Honesty Score	0-1	Research integrity rating. 0.8+ is good
Confidence	0-1	How confident is the AI in this result

 How Results Are Generated & Stored

Step 1: Upload Papers

```
You select PDF files
|
└─> Backend receives papers
    └─> Converts to JSON with title, abstract, content
```

Step 2: Send to N8N

```
Backend sends papers to N8N webhook
|
└─> N8N receives them
    └─> Creates execution record
```

Step 3: Run All Modules

```
N8N runs 10 modules in orchestrated workflow
|
├─> M1 summarizes each paper
├─> M2 groups papers into topics
├─> M3 finds gaps between topics
├─> M4 tracks trends
├─> M5 creates visualization
├─> M6 finds bridge papers
├─> M7 detects contradictions
├─> M8 builds method-dataset matrix
├─> M9 generates related work
└─> M10 calculates honesty score
```

Step 4: Return Results

```
N8N formats all results
|
└─> Sends back to backend
    └─> Backend saves to MongoDB
        └─> Returns to frontend
            └─> Frontend displays 9 sections
```

Step 5: Save & History

```
Each analysis is saved with:
- Timestamp
- Paper count
- Topic count
- Gap count
```

- Year range
- Quality score
- All module results

This is stored in AnalysisRun in MongoDB
and IndexedDB in browser for offline access

Example: Real Analysis Walkthrough

Let's say you upload 100 AI research papers from 2020-2024:

Module 1 Summary: "Found 100 papers. Created summaries for all."

Module 2 Topics: "Found 12 topics:

- Deep Learning (28 papers)
- NLP (22 papers)
- Computer Vision (18 papers)
- Knowledge Graphs (15 papers)
- Reinforcement Learning (12 papers)
- ... (7 more)"

Module 3 Gaps: "Found 34 significant gaps:

- #1: Deep Learning & Knowledge Graphs (score: 0.85) └ Similar (0.72) but never appear together (0)
- #2: NLP & Computer Vision (score: 0.78) └ Similar (0.68) but rarely together (1 paper)
- #3: Reinforcement Learning & Knowledge Graphs (score: 0.72) ... (31 more)"

Module 4 Trends: "Deep Learning is RISING (↗)

- 2020: 8 papers
- 2021: 15 papers
- 2022: 22 papers
- 2023: 28 papers
- 2024: 35 papers Growth rate: 45% per year"

Module 5 Map: "Created visualization with:

- 100 points (papers)
- 12 colored clusters (topics)
- 34 curved lines (gaps between topics)"

Module 10 Honesty: "Average honesty score: 0.87

- Most papers cite relevant work
- Few contradictions found
- 2 papers have citation gaps"

Dashboard Result:

Section 1: 100 papers, 12 topics, 34 gaps, 2020-2024
Section 2: Summary of key findings
Section 3: Top 10 gaps with explanations
Section 4: Trend analysis (Deep Learning rising)
Section 5: Interactive research map
Section 6: Q&A about findings
Section 7: Research integrity metrics
Section 8: Which methods tested on which datasets
Section 9: Auto-generated related work section

Tips for Interpreting Results

Look For These Patterns:

1. High Gap Scores with 0 Co-occurrence

- Indicates unexplored connection between similar topics
- Best opportunity for new research

2. Rising Trends

- "Hot" research areas
- Good for staying current
- May be competitive

3. High Coherence (0.8+)

- Well-defined topics
- Clear research direction
- Easier to understand

4. Low Similarity with High Co-occurrence

- Disparate topics that work together
- May indicate emerging interdisciplinary field

Watch Out For:

1. Low Coherence (< 0.6)

- Topic may be too broad or poorly defined
- Review papers in that topic manually

2. Very Few Papers in a Topic

- May not be a significant area
- Treat with caution

3. Zero-occurrence Gaps

- Could be genuine opportunities OR
- Topics may be conceptually incompatible

4. Declining Trends

- Topic may be mature (no new problems)
- Or replaced by newer approaches



Using Results for Your Research

Finding Research Questions:

1. Identify high-scoring gaps (Section 3)
2. Find papers that mention both topics (click gap)
3. Ask: "Why haven't these topics been connected?"
4. Propose: "What if we combined them?"

Literature Review:

1. Go to Section 9 (Related Work Draft)
2. Copy the auto-generated text
3. Verify accuracy (always check AI output!)
4. Expand with your own insights
5. Cite properly

Identifying Trends:

1. Check Section 4 (Trend Analysis)
2. Find rising topics in your area
3. Review papers from rising topics
4. Position your work within the trend

Spotting Contradictions:

1. Check Section 7 (Scientific Honesty)
2. Review contradictions found
3. Investigate why different results
4. Resolve in your own research



Technical Details for Developers

Result Structure (Saved in MongoDB)

```
{
  _id: ObjectId,
  userId: "user_123",
  timestamp: "2024-05-15T10:30:00Z",

  // Summary
  papers: 145,
  topics: 12,
  gaps: 34,
  qualityScore: 0.82,
  yearRange: { start: 2020, end: 2024 },

  // Full results from all modules
  modules: {
    module1: { summaries: [...] },
    module2: { topics: [...], assignments: [...] },
    module3: { gaps: [...] },
    module4: { trends: [...] },
    module5: { map: { points: [...], links: [...] } },
    module6: { gapEvidences: [...] },
    module7: { contradictions: [...] },
    module8: { matrix: {...} },
    module9: { relatedWork: {...} },
    module10: { honestyScore: 0.87, ... }
  },

  // Analysis report
  analysisReport: {
    reportTitle: "Analysis Report",
    reportSummary: "...",
    reportMarkdown: "...",
    executive_summary: "..."
  },

  // Metadata
  status: "completed",
  processingTime: 135000, // milliseconds
  n8nExecutionId: "exec_123"
}
```

Frontend Display Flow

```
ResultsSection.tsx
├─ Section 1: DatasetSummarySection
├─ Section 2: ReportSummarySection
├─ Section 3: GapDetectionSection
├─ Section 4: TrendAnalysisSection
├─ Section 5: ResearchMapResultsSection
├─ Section 6: ChatResultsSection
├─ Section 7: EvaluationSummarySection
```

- └ Section 8: DatasetMethodMatrixSection
- └ Section 9: RelatedWorkDraftSection

Each section:

1. Extracts data from backendResult.modules
2. Transforms for display (normalization, filtering)
3. Renders interactive UI
4. Handles user interactions (clicks, hovers)

Mobile & Print

Print PDF:

- Click "Print" button in top-right
- Selects content to print
- Generates HTML with all sections
- **Includes:** Analysis Summary + Quality Metrics + All sections
- Opens print dialog
- Can save as PDF

History:

- All analyses saved in History tab (sidebar)
- Shows timestamp, paper count, topic count, gaps
- Click to view previous analysis
- Delete individual analyses
- Clear all history

Summary

ResearchLens is a complete research analysis system that:

1. **Reads** papers (Module 1)
2. **Understands** them by finding topics (Module 2)
3. **Connects** topics to find gaps (Module 3)
4. **Tracks** how topics evolve (Module 4)
5. **Visualizes** relationships (Module 5)
6. **Finds bridges** for gaps (Module 6)
7. **Detects issues** (Modules 7, 8, 10)
8. **Writes reports** (Module 9)
9. **Displays everything** in 9 interactive sections

N8N orchestrates all modules to run efficiently and reliably.

The result? A comprehensive understanding of any research domain in minutes instead of weeks!

Related Documents

- [N8N_INTEGRATION_SUMMARY.md](#) - How N8N is configured
 - [N8N_WORKFLOW_QUICK_REFERENCE.md](#) - N8N workflow templates
 - [N8N_SETUP_GUIDE.md](#) - How to set up N8N
 - [README.md](#) - Project overview
 - [WORKFLOWS_GUIDE.md](#) - Workflow usage guide
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